

Extended Essay

May 2025 Session

To what extent can the Monte Carlo Circle method of calculating π provide a more accurate and precise approximation of π compared to Buffon's Needle method?

Word Count: 3995

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1. Introduction

Pi (π) is an essential component in the development of mathematics. Originating from the Babylonians in 1900 BCE (Exploratorium), the π value of 3 introduced the concept of a ratio: the ratio of circumference over diameter of a circular shape. This irrational value is now precisely known to 105 Trillion digits as of June 2024 (Baker).

Though there are a wide variety of ways to approximate π , probability analysis is the most fascinating for me, because I had never thought that a series of seemingly random experiments would be capable of calculating such a unique number. This investigation will focus on two methods of calculating π : *Buffon's Needle*, and what is known as the *Monte Carlo Circle Method*. Both methods are based on the "Monte Carlo Simulation" which is a probabilistic method of calculating π by using computer generated random numbers as data values in the calculations.

The first "Monte Carlo" simulation occurred after World War II where Stanislaw Ulan, a mathematician, coined the term "Monte Carlo" after the *Monte Carlo Casino*, the world symbol of gambling as Las Vegas is to us today (Explained: Monte Carlo Simulations). Nowadays, there is widespread usage of Monte Carlo Simulations in estimation problems.

The goal of this investigation is to determine both the accuracy and precision of the two probabilistic methods for calculating π . This will be done by performing experiments using the Monte Carlo Method for both *Buffon's Needle Method* and *Monte Carlo's Circle Method*. A simulation will be run repeatedly and tested using repeated trials to reduce variability. Then, the data obtained will be compared by using standard error,

root mean squared error, and other comparison methods. Analyzing both these perspectives in combination provides the most comprehensive understanding of calculating π because we want both a high precision and high accuracy in calculations. The contrasting elements between these two methods will reveal the effectiveness of such probabilistic methods and whether they can be applied in real life to calculate π .

The investigation will provide a thorough analysis of both methods and compare them in order to gain understanding of the practicality of π .

In other words, can random events always end in concrete results?

2. Monte Carlo's Methods

As has stated, Monte Carlo Methods use random data sets, which are often computer generated. Thus, "The effectiveness of Monte-Carlo simulation depends crucially on the use of good pseudo-random number generators" (Stevens). Computers use pseudo-random number generation because software simply cannot produce true random numbers, which don't rely on any sort of algorithm. Specifically, there is a definite sequence to all these generated numbers even though they look quite random to us (True Random vs. Pseudo Random Number Generation). In order for this investigation to be most accurate, it must be ensured to the greatest extent possible that random-enough data is generated.

The law of large numbers is a fundamental theory in probability and statistics. It is a generic name given to a group of theorems describing how experimental values are the closest to theoretical with ever increasing sample sizes (Dunn). The law states that "The average of the results obtained from a large number of independent random samples

converges to the true value, if it exists” (Earl and Nicholson 234). The weak law of large numbers is most applicable to calculating π using Monte Carlo’s methods because it suggests convergence based on probability. As sample sizes increase, the probability of reaching the theoretical value increases along with it, but without certainty. The Gambler’s Fallacy provides an interesting perspective on this law. Take a coin toss for example, if it has landed on tails five times in a row, one might think the next toss has a higher chance of landing on heads. This seems like a reasonable assumption until you consider that the coin has no memory of past trials, which explains why in a casino, although short term odds vary greatly, the house always wins in the long term (The Law of Large Numbers and the Gambler’s Fallacy). As this law is the essence of Monte Carlo’s methods, it is unquestionable that the name of these methods comes from a casino. This is a law that has puzzled gamblers and non-gamblers alike since its introduction in the 17th century (Barbu and Zhu).

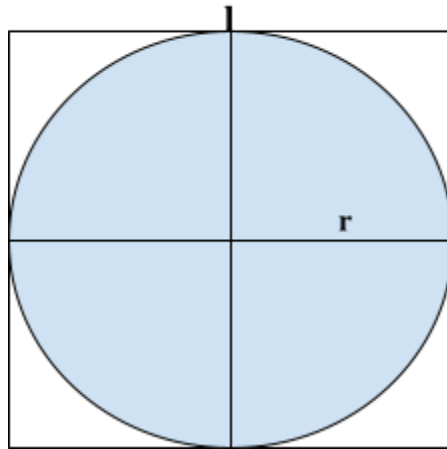
Below is the equation for the law of large numbers, where “n” represents the number of data points in the set. This is usually a very large number.

$$\frac{Sn}{n} = \frac{1}{n} (X_1 + X_2 + \dots + X_n)$$

3. Monte Carlo’s Circle Method

Monte Carlo’s “Circle method”, is the simplest probabilistic method of calculating π . This one was pretty simple for me to understand as the simulation is easy to recreate.

This method relies on a generator to plot random points in the diagram drawn below, with a circle inscribed within a square. The radius of the circle is r , and the length of the circle is twice the radius (Easterday and Smith). One way to overcome the pseudorandom generator is to perform the experiment away from a computer, such as dropping rice onto a grid myself (Williamson). However, a lot of human error such as miscounting (judgment error), or the fact that each piece could interfere with the dropping of another, as well as just how tedious the experiment would be uproots the accuracy of such an experiment.



$$l = 2r$$

$$\text{Area of Circle} = \pi r^2$$

$$\text{Area of Square} = l^2 = (2r)^2 = 4r^2$$

$$\frac{\text{Area of Circle}}{\text{Area of Square}} = \frac{\pi r^2}{4r^2} = \frac{\pi}{4}$$

Imagine that the density of points is so large that all the space in the diagram is filled up. The ratio of the points that land within the circle to the total number of points in the square, would be equal to the ratio of their areas, which corresponds to $\pi/4$. This explains why more points provides a more accurate estimation; because more area is “covered”.

This can be rearranged to solve for π .

$$\pi = \frac{4(\# \text{ Points in Circle})}{(\text{Total } \# \text{ Points})}$$

I initially created my own simulation of the Monte Carlo Circle using python. However, I realized as soon as I finished that performing experiments using it is unrealistic because generating millions of data points took up to an hour to finish. Even then, the accuracy I was getting was quite low and not suitable for the experiment, so I found an online simulation instead.

The essential mathematical concept behind the simulations for the circle is to apply the equation of a circle to determine if a point falls within the confines of the circle.

$$x^2 + y^2 = r^2$$

If this expression holds true for the coordinates of a point (x, y), with r being the radius of the circle in the simulation, then that point falls within the circle.

$$x^2 + y^2 < r^2$$

If they happen to be equal, then it is right on the circle's edge, and to count or not it would be determined by whether or not a point on the border or either shape would be counted. This was my big question with both my simulation, and with the one I found online: Is the circle border inclusive or exclusive? If a plotted point lands perfectly on the edge of the circle or the edge of the square, is it counted as "in"?

I did some research and found that most simulations and experiments include any point that lands exactly on the circle's circumference as "in" the circle. However, from both a statistical and probabilistic perspective, whether or not we consider the points that land on the circumference will not skew the results, especially when sample sizes increase. If we compare the perimeter of the square to the perimeter of the circle, we find that there is a $4/\pi$ higher chance of a point landing on the square's perimeter than the circle. This is a small probability by itself, and, behind the simulation, very precise coordinates are plotted which makes the probability of one landing on the edge extremely low.

Table 1. Experimental Data Obtained Using a Simulation from *Engaging Data*, A Javascript Model for Monte Carlo's Circle Method

	1000	10000	100000	1000000
Trial 1	3.06800	3.11800	3.13292	3.14300
Trial 2	3.08000	3.14040	3.15048	3.14277

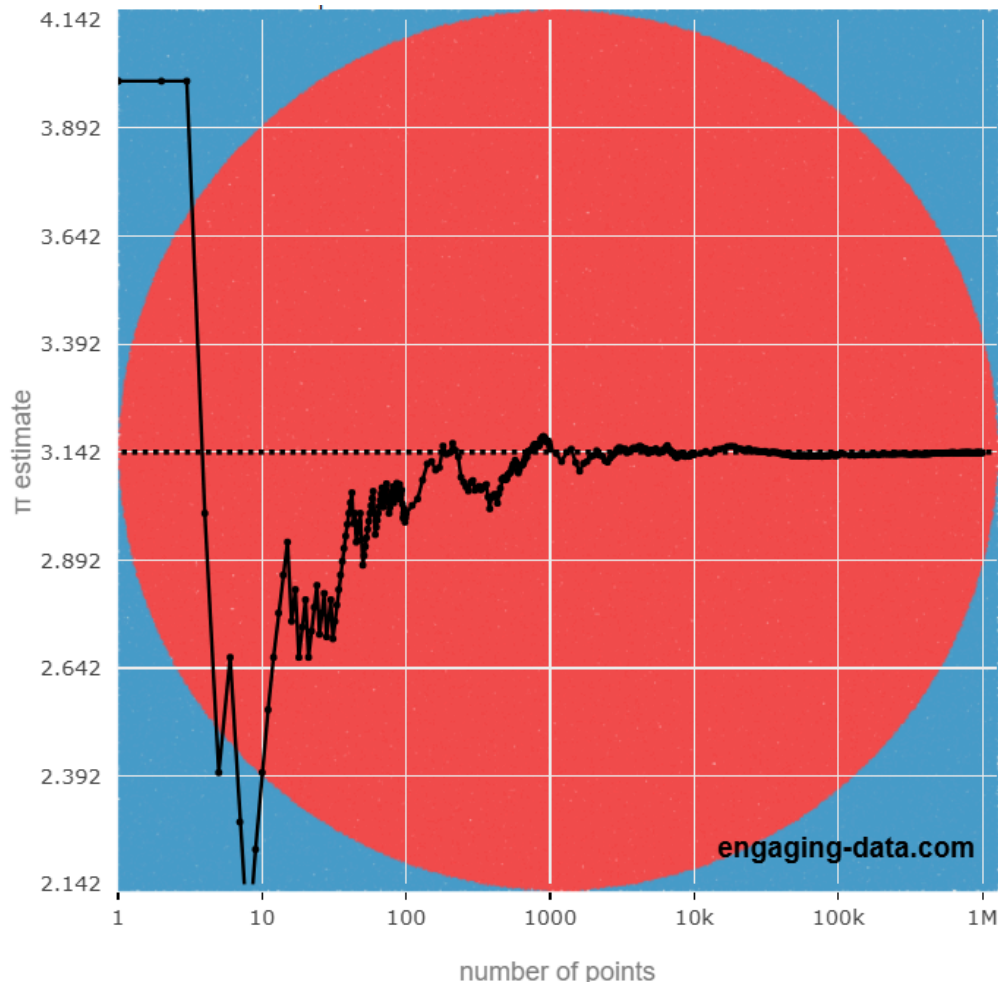
Trial 3	3.13200	3.11560	3.13688	3.13950
Trial 4	3.15600	3.13920	3.14208	3.14371
Trial 5	3.14800	3.15760	3.15316	3.14388
Trial 6	3.18400	3.15760	3.14520	3.14034
Average	3.12800	3.13807	3.14345	3.14220
Range	0.11600	0.04200	0.02024	0.00437
Difference from π	0.01359	0.00353	0.00186	0.00061

**The values are rounded to 5 decimal places, which seems fit for the precision of data obtained.*

The graph below the graph that has been created from the simulation data. It shows the number of points plotted on the x-axis and the value of π calculated on the y-axis.

Fluctuations are evident where the black line graph goes above and below the dotted line, representing the calculated “ π ”, as it converges closer to the true value. The trend shows that, with larger data sets, the calculated value becomes more accurate. I ran the simulator multiple times and each time it ended with a nearly horizontal trend line, showing the consistency of this method.

Figure 1: Graph of Monte Carlo’s Circle Method results converging to π from *Engaging Data, A Javascript Model for Monte Carlo’s Circle Method*



Relating back to the law of large numbers, a very small sample of 5 points could easily have five points that all lay outside of the circle, or all lie inside it. An increase in the number of points would reduce this likeness, even though something could theoretically happen with a large number of points.

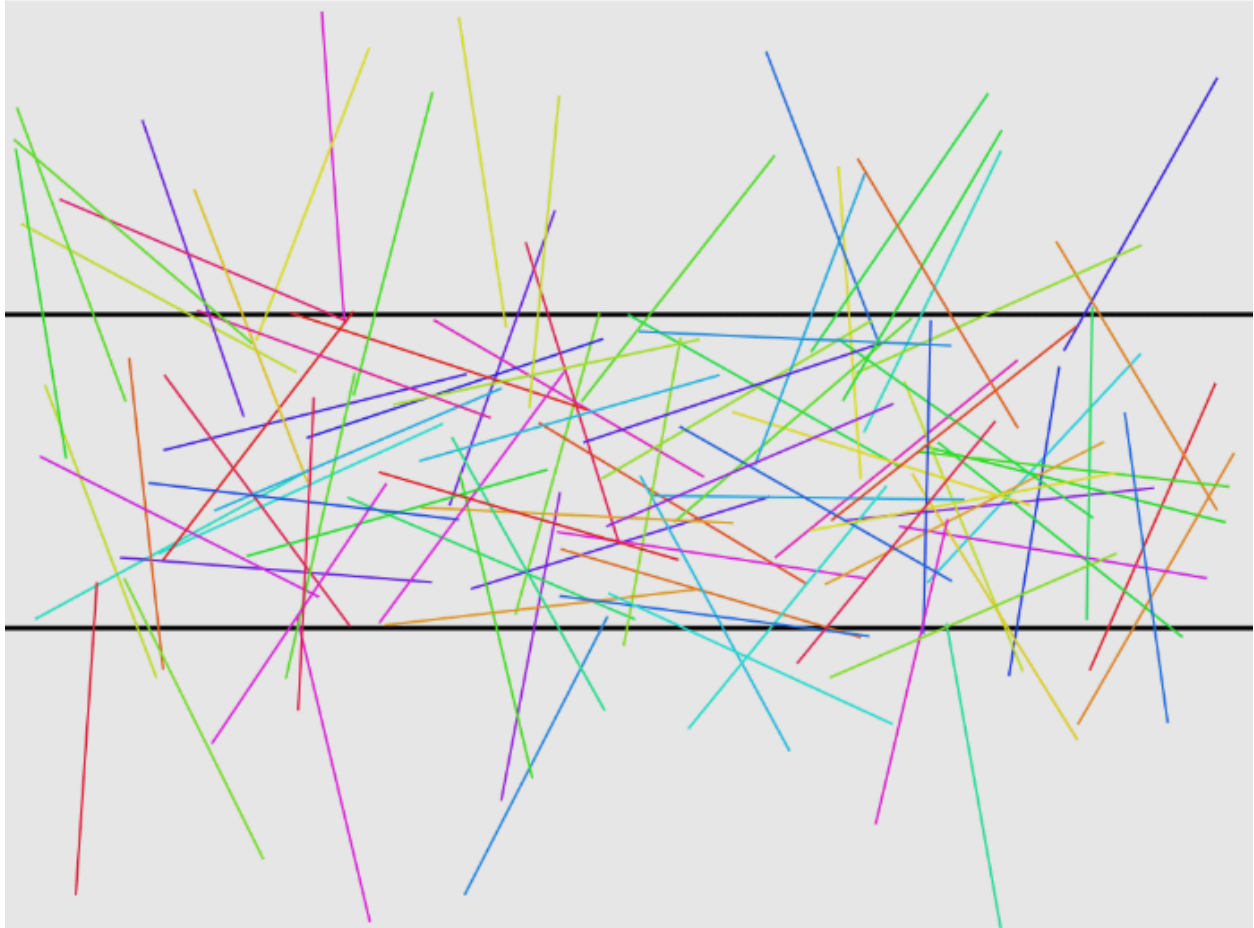
4. Buffon's Needle Method

My initial impression of this method was one of surprise. Compared to the Monte Carlo Circle, this one is infinitely more abstract and harder to understand, as I didn't think natural occurring objects could somehow model π . In contrast to the circle method, this

experiment is much more difficult to recreate as it requires many factors such as width and length of the needle, and the angle at which it lands. Doing this experiment physically also has many more errors than the circle method because using physical needles would require quite a bit of space as well as the fact that the weight and density of needles are not evenly distributed. The spread of the “board” would make it difficult for us to randomly drop very many needles in random places because of human error. For this experiment, we will have to assume that the randomizer is random enough to produce results, as I cannot reproduce the experimental results to act as a reference.

Buffon's Needle is commonly known as the earliest “Monte Carlo” simulation, and is considered the first ever experiment of geometric probability. Georges-Louis Leclerc, Comte de Buffon posed and solved this problem himself (Velasco). Using an experiment that involves randomly tossing needles onto a uniformly lined surface with the length between two lines as d . We will assume that the length of the needle, l , is less than d (McRae).

Here is an example where 100 needles were dropped, using colored “needles” to differentiate the lines from the grid (Buffon's Needle, MSTE, University of Illinois).



The probability of a needle intersecting the line (Expected value) was proved by Buffon to be $2l/\pi d$.

$$E(X) = 0(\text{Probability that } X = 0) + 1(\text{Probability that } X = 1)$$

The expected value is calculated by setting a random variable X to equal 0 if no intersection occurs and setting it equal to 1 when an intersection does occur. This gives us the probability of an intersection at all.

$E(X)$ is dependent on only the length, so the expected value function can be modeled like this:

$$E(X) = f(l)$$

As the length of the needle increases, the expected value must also increase. Logically, a larger object would have a greater chance of intersecting a line.

Assuming dropping one needle doesn't interfere with the dropping of another, then dropping multiple needles at once has the exact same probability as dropping them one after the other. Thus, it can be proved that the expected values for multiple needles summed together is the same as the expected value for each probability summed up.

$$E(X_1 + X_2 \dots) = f(l_1 + l_2 \dots)$$

$$E(X_1) + E(X_2) \dots = f(l_1) + f(l_2) \dots$$

$$f(l_1 + l_2 \dots) = f(l_1) + f(l_2) \dots$$

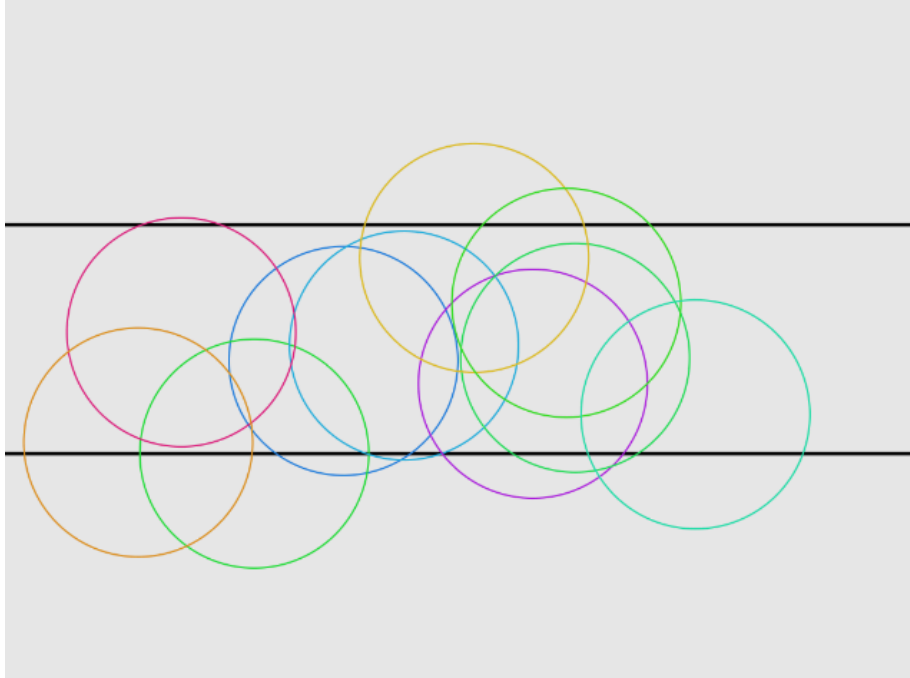
Equating the two equations established prior shows that the function $f(l)$ must be linear in order for the third line to be true.

$$E(X) = Nl$$

Where N is a constant of the linear equation, because linear equations can be manipulated to appear in slope intercept form

$$y = mx + b$$

This is what will be discovered as equating to $2/\pi d$.



On another note, the expected value if this experiment is done with circles is always 2, if the circle has a diameter that is the same length as the distance between two boards, because the circle will always intersect two lines perpendicularly, or the same line twice any other time.

Thus, this is the expression obtained for the expected value using the circles:

$$E(X) = Nl = 2$$

$$N = \frac{2}{l} = \frac{2}{\pi d}$$

$$E(X) = \frac{2l}{\pi d}$$

Table 2. Experimental Data for Buffon's Needle Obtained Using a Simulation from University of Illinois Urbana-Champaign

	1000	10000	100000	1000000
Trial 1	3.15956	3.15408	3.13229	3.14229
Trial 2	3.18979	3.13725	3.12632	3.13985
Trial 3	3.12500	3.11333	3.16196	3.13969
Trial 4	3.23625	3.17309	3.14292	3.13873
Trial 5	3.17460	3.09933	3.14693	3.14208
Trial 6	3.01205	3.11915	3.13612	3.14263
Average	3.14954	3.13271	3.14109	3.14088
Range	0.22420	0.07376	0.03564	0.00391
Difference of average from π	0.00795	0.00889	0.00050	0.00072

5. Comparison

After collecting data using both the experiments above, the accuracy and precision can be compared. The easiest way of comparing accuracy for each experiment is to find the average of the difference between the calculated numbers for π at each manipulation, and the average of the range. A smaller average difference from π means the results produced are more accurate, whereas a smaller average range means the results are more precise.

$$\text{Average Difference (Circle)} = \frac{0.01359 + 0.00359 + 0.00186 + 0.00061}{4} = 0.00491$$

$$\text{Average Difference (Needle)} = \frac{0.00795 + 0.00889 + 0.00050 + 0.00072}{4} = 0.00452$$

Comparing this way, we see that Buffon's needle has a lower difference from π , meaning it is more accurate. However, both 0.00491 and 0.00452 are rather large

relative to how precise π itself is, so these models are not the greatest to calculate π to more than a certain number of decimal places.

What's easier to visualize for a general measure of accuracy is the "error" as a percentage: Percent Error. To get a comprehensive experimental "result" for each experiment, we will be averaging the results of the multiple trials above to give us an experimental value to put in this equation. This is found by taking the absolute value of the difference between experimental and actual values over the actual value and multiplying by 100% as shown below.

$$\text{Percent Error} = \left| \frac{\text{Theoretical} - \text{Experimental}}{\text{Theoretical}} \right| \cdot 100\%$$

The experimental value used is the average of the four trial's average values of π calculated, and the theoretical value used is 3.14159, both to 5 decimal places.

Sample Calculation (Monte Carlo Circle):

$$\text{Average of Averages (Table 1, Row 8)} = \frac{3.12800 + 3.13807 + 3.14345 + 3.14220}{4} = 3.13793$$

$$\text{Percent Error} = \left| \frac{3.14159 - 3.13793}{3.14159} \right| \cdot 100\%$$

$$= 0.11650\%$$

Doing the same calculation with Buffon's Needle Data gets this result:

$$= 0.01687\%$$

Using this general test for experimental data, we see that Buffon's experiment is more accurate in calculating π because it only deviates from the actual value by an error percentage of 0.01687%, which is only about 0.05 whereas the Monte Carlo's Circle Method has a percent error of 0.11650%, significantly higher and a whole 0.37 numerical error from the actual π value of 3.14159. Since neither of these experiments have extremely low percent error, it is not effective to use the theoretical value as a more precise value.

In terms of precision, it is useful to look at the range of data obtained. A smaller range means data is more concentrated, whereas a large one means data is spread out, and thus less precise. Average range will be calculated using the data from Row 9 from Tables 1 and 2 above.

$$\text{Average Range (Circle)} = \frac{0.11600 + 0.04200 + 0.02024 + 0.00437}{4} = 0.04565$$

$$\text{Average Range (Needle)} = \frac{0.22420 + 0.07376 + 0.03564 + 0.00391}{4} = 0.08438$$

Though Buffon's Needle is the most accurate experiment according to the analysis above, its range is nearly double that of Monte Carlo's Circle Method. This means that the values are much more imprecise and that more variability could occur with larger sets of data.

Next, Mean Squared Error will be used to compare the two experiments. This is the average squared difference between the estimated values and the actual values (Earl and Nicholson 261). Mean squared error measures the average of the squares of the

errors. This means that the magnitude at which a value deviates from π will be squared and added together. Mean squared error ensures that positive (greater) errors do not cancel out negative (lower) ones. The average of these squared errors provides a single measure of model accuracy. A low error suggests a good fit (the model's predictions are close to actual values) and a larger one suggests that the model is not suitable for this calculation.

Taking the square root of mean squared error gives you root mean squared error, or RMSE. This makes the values of error obtained much easier to understand because it relates directly to the scale of the data.

$$MSE = \frac{1}{m} \sum_{n=1}^m (\pi - x)^2$$
$$RMSE = \sqrt{\frac{1}{m} \sum_{n=1}^m (\pi - x)^2}$$

I want to measure the RMSE for the entire experiment at large, rather than each manipulation independently. To do this, the RMSE values for each manipulation are averaged to find a total RMSE value for each experiment to compare. This functions on the assumption that the errors for each manipulation are of equal “weight” (Even though the trend in error size generally decreases as sample sizes increase, we assume that each one is equally likely to happen, and considered equally in calculations). If it was possible to do even larger sample sizes, then it would be better to graph out the “errors”

as a function of sample size and see the graph converge to zero, but this is not something that can be done with the data I have available.

Table 3: Processed data for Calculating RMSE using the Circle Method

	Absolute value difference from π , using π rounded to five decimal places					
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6
1000	0.07359	0.06159	0.00959	-0.01441	-0.00641	-0.04241
10000	0.02359	0.00119	0.02599	0.00239	-0.01601	-0.01601
100000	0.00867	-0.00889	0.00471	-0.00049	-0.01157	-0.00361
1000000	-0.00141	-0.00118	0.00209	-0.00212	-0.00229	0.00125

Sample Calculation for RMSE (Circle Method for 1000 points):

$$RMSE (1000 Points) = \sqrt{\frac{(0.07359)^2 + (0.06159)^2 + (0.00959)^2 + (-0.01441)^2 + (-0.00641)^2 + (-0.04241)^2}{6}}$$

0.04348970874

Average of all RMSE calculated for each manipulation.

$$Average RMSE = \frac{0.04348970874 + 0.01708672292 + 0.007343048867 + 0.001781834261}{4}$$

0.0174253287

Table 4: Processed data for Calculating RMSE using Buffon's Method

	Absolute value difference from π , using π rounded to five decimal places					
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6
1000	-0.01797	-0.04820	0.01659	-0.09466	-0.03301	0.12954
10000	-0.01249	0.00434	0.02826	-0.03150	0.04226	0.02244

100000	0.0093	0.01527	-0.02037	-0.00133	-0.00534	0.00547
1000000	-0.00070	0.00174	0.0019	0.00286	-0.00049	-0.00104

$$\text{Average RMSE} = \frac{0.07041800942 + 0.02663072943 + 0.01150945408 + 0.001664777262}{4}$$

$$= 0.02755574255$$

Buffon's method produces a higher RMSE than Monte Carlo's Circle, meaning the error from the true value of π is greater, thus meaning this method is less accurate than the circle method if compared using this model.

This is similar to standard deviation, the second method of comparison, but where standard deviation measures error from the average value, RMSE measures deviation from the "true" value. This means that RMSE truly measures "error", focusing on accuracy over precision, whereas standard deviation is only a measure of how spread out the values are and focuses on precision over accuracy.

The formula for standard deviation is listed below: It is the sum of the differences between values and the mean, squared, and divided by m, the number of values.

$$\sigma = \sqrt{\frac{1}{m} \sum_{n=1}^m (\mu - x)^2}$$

Table 5: Processed data for Calculating Standard Deviation using the Circle

Method

	Average s	Difference from average (using averages in column 2 rounded to five decimal places)
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		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6
1000	3.12800	0.06000	0.04800	-0.00400	-0.02800	-0.02000	-0.05600
10000	3.13807	0.02007	-0.00233	0.02247	-0.0113	-0.01953	-0.01953
100000	3.14345	0.01053	-0.00703	0.00657	0.00137	-0.00971	-0.00175
1000000	3.14220	-0.00080	-0.00057	0.00270	-0.00151	-0.00168	0.00186

Sample Calculation for Standard Deviation (Circle Method for 1000 points):

$$\sigma = \sqrt{\frac{(0.06)^2 + (0.048)^2 + (-0.004)^2 + (-0.028)^2 + (-0.02)^2 + (-0.056)^2}{6}}$$

$$= 0.04131182236$$

$$\text{Average } \sigma = \frac{0.0413118224 + 0.0173381396 + 0.0071027014 + 0.0016741665}{4} = 0.01685670747$$

Table 6: Processed data for Calculating Standard Deviation using the Buffon's Method

	Average s	Difference from average (using averages in column 2 rounded to five decimal places)					
		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6
1000	3.14954	-0.01002	-0.04025	0.02454	-0.08671	-0.02506	0.13749
10000	3.13271	-0.02137	-0.00454	0.01938	-0.04038	0.03338	0.01356
100000	3.14109	0.00880	0.01477	-0.02087	-0.00183	-0.00584	0.00497
1000000	3.14088	-0.00141	0.00103	0.00119	0.00215	-0.00120	-0.00175

$$\text{Average } \sigma = \frac{0.0699676143 + 0.0251048312 + 0.0610879725 + 0.0015049972}{4} = 0.0394163538$$

The Central Limit Theorem (CLT) connects the distribution of a growing sample from a given distribution with the normal distribution. (Dunn) It states that “the distribution of sample means approaches a normal distribution as the sample size increases, regardless of the shape of the population distribution.” (Earl and Nicholson 55) This

means that in order for my data to truly approximate a normal curve, I would need a large number of trials to be completed in order to determine if these experimental results resemble a Gaussian distribution characterized by the symmetrical bell-shaped curve. A Gaussian Distribution is a continuous probability distribution curve that is widely used in probability theory because of its many applications. This curve can be used to visualize mean and standard deviation, and due to its ability to be transformed into a standard normal curve using z-score, it is an easy way to visually compare two probabilistic samples. Standard normal curves can use a sample to represent the population. For a distribution to be Gaussian, it has to have a mean and standard deviation where almost all the population lies within 3 standard deviations of the mean. Nearly every single data point from the original data in Tables 1 and 2 exist within one standard deviation from the mean, and essentially all the data within 3 standard deviations (with the rare exception of outliers). This is a good sign, showing that the data closely approximates a normal distribution, making it quite precise. The standard deviation of Buffon's experiment is more than double that of the Circle experiment. This means that the latter is more precise and able to generate values of π within a certain window, lowering the possibility of outliers. However, this experiment lacks accuracy when compared to Buffon's needle.

The last method of comparing is another measure of variability of the data obtained: Standard Error. Standard Error can tell us how useful each simulation would be if a large sample were to be tested (Earl and Nicholson 393). It measures how likely the sample mean is to differ from the population mean. It is also important because rather

than describing the variability of data within one manipulation (to then be averaged afterwards), Standard error can estimate the variability across multiple samples. The limitation of this is that the standard deviation that is required in the equation comes from a sample, and not the population.

Standard Error, or Standard Error of the mean (SE) is calculated by dividing the standard deviation of the sample by the number of elements in the sample, n .

$$SE = \frac{\sigma}{\sqrt{n}}$$

$$SE(Circle) = \frac{0.04131182236}{\sqrt{4}} = 0.02065591118$$

$$SE(Needle) = \frac{0.0394163538}{\sqrt{4}} = 0.0197081769$$

Standard error is commonly represented alongside the mean like an uncertainty.

$$Circle: 3.13793 \pm 0.02066$$

$$Needle: 3.14106 \pm 0.01970$$

Since this comparison method relies on sample standard deviation, which is calculated from raw data and then put into the standard error equation, the effectiveness of this comparison is greatly reduced. Thus, this comparison is best used only as a reinforcement to other comparison methods, and should not be relied on.

6. Conclusion

Through comparison methods like percent error, standard error, standard deviation, and root mean squared error, we come to the conclusion that Buffon's experiment produces results that are more accurate than Monte Carlo's Circle. This method also requires more computer space to run, and, in a real-life situation, is harder to perfect. These results are comparably less precise than what is generated through the Circle method. There are positive aspects to each experiment, there are also challenges with both. The Monte Carlo Circle method is one of the easiest ways to calculate π , and makes the most sense in terms of how it works to calculate the real value. I believe this simplicity makes it stand out more despite being less accurate than Buffon's needle, which is more abstract and harder to recreate in real life.

Taking the accuracy and precision and efficiency of each of these experiments into account, and from the results gathered in this investigation, it is evident that neither Monte Carlo's circle method, nor Buffon's needle method is suitable for calculating π in a mathematical sense, because neither is capable of producing more than 3 digits of accuracy even with thousands of data points. However, we can see that Buffon's Needle, although significantly less accurate with small amounts of data, is much more accurate and precise in comparison to the circle method as the data sample increases in size. For precise mathematical calculations, I would not recommend either of these methods as probability, even at its best, is still "random". The accuracy and precision of results produced is not favored, and would only be useful to look at if you want a simple

representation of applying different methods to calculate π . For results to be truly useful, I would instead apply infinite series to calculate π , as these are more reliable because of their fixed structure. The Ramanujan and Leibniz infinite series are commonly used to achieve record breaking calculations of π .

There are a number of improvements that could be made to each experiment for more accuracy and precision. Something that is evident from the comparison is that the amount of data I have collected is far too limited. The best improvement would be to perform multiple more trials with a broader range of numbers of data points. This is because the entire point of sampling is to mimic the population, but there is always sampling error. Using a large sample size would be the best solution to minimizing this error. Another improvement would be to perform the experiments using a bigger radius for a circle, or a larger sample space for Buffon's needle, which would mean data points are generated with more breadth, giving me a broader spread of results.

A look into both of these experiments allows individuals to further understand the mathematics of calculating π . This investigation has shown me what π really is, and opened me up to new areas of mathematics, like the Gambler's Fallacy, and other methods of calculating π which would be very insightful to research in the future.

While researching for this investigation, nearly every source brought me back to the concept of a circle. From the circle inscribed within the square, to the circle drawn around the needle in Buffon's method, the repetitive usage of circles to explain π shows

its connection to geometry in everyday life, and why it is essential in modeling π in both the mathematical world and the real world. I would agree with the statement that random events can always end in concrete results, because there is always some underlying order to it. Both these seemingly probabilistic ideas require circles, which directly relate to π , or at least, they show us how closely π is correlated to circles. Thus, because fixed circles are present, the experiment itself isn't as random as it seems, and neither are the results. The same idea is present with the idea of pseudo random numbers, which I have used to generate data. There is a hidden algorithm underlying this, that really makes this randomness, not so random.

The possibilities of such a mysterious number are boundless. From probability, to infinite series, to geometric shapes and simple ratios, π is a number that exists everywhere we look. We can only imagine what this number will lead us to discover in the future.

7. Bibliography

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